Roberto Barreda

Senior Software Engineer | Cloud Native | Technical Leadership

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Summary

15+ years building cloud-native distributed systems and leading technical teams. Led teams of 3-8 engineers on products used by millions, with 99.95%+ uptime. Focus on zero-trust security, observability, and edge AI deployment.

Core Competencies

- **Technical Leadership** — Lead engineering teams and make architecture decisions
- **Product Mindset** — Connect engineering work to product goals
- **Cloud Architecture** — Design AWS, GCP, and Azure systems with Kubernetes
- **System Reliability** — Use metrics and tracing to keep systems available
- **Cross-functional Collaboration** — Work with product, operations, and support teams
- **Team Mentorship** — Coach engineers and help them grow

Work Experience

Senior Software Engineer - Technical Lead — Midokura (Sony Group)

January 2024 – Present • Remote

Lead a team responsible for inference infrastructure for AI workloads in private data centers. Previously led services for Sony's AITRIOS platform. Managed smart edge devices and deployed third-party AI models and WASM applications at scale.

Key Achievements:

- Built Go services and Kubernetes orchestration that provision GPU hardware and run inference workloads
- Built vLLM inference services with model selection, tuning, semantic routing, packaging, deployment, and observability
- Integrated 50+ AI models and deployed WASM runtimes on Kubernetes across 10K+ edge devices. Cut model onboarding from weeks to days, increased adoption by 40%, and reached 99.9% deployment success
- Established CI/CD and automated testing, 3x deployment frequency and cutting production incidents by 60%
- Led design reviews and mentored 4 engineers in cloud-native architecture and edge computing

Technologies: Java, Python, Go, vLLM, GPU infrastructure, Kubernetes, Azure, Terraform, SQL, Prometheus, GitOps, IaC

Staff Software Engineer — Netskope

October 2021 – January 2024 • Remote

Built zero-trust security features and highly available microservices for the Remote Browser Isolation (RBI) platform serving 5M+ users.

Key Achievements:

- Architected an orchestration and configuration engine that managed 1K+ browser instances and processed 50K+ requests/hour with sub-100ms latency
- Raised uptime from 99.5% to 99.95% through circuit breakers, automated failover, and chaos engineering
- Built an observability platform with Prometheus, Grafana, and distributed tracing. Cut MTTR by 70% and enabled proactive incident response
- Established GitOps workflows and Terraform/Helm IaC, enabling 50+ deployments per week with zero-downtime releases
- Mentored 5 engineers in microservices, Kubernetes, and security-first design

Technologies: Go, Python, Kubernetes, Google Cloud (GCS), Prometheus

Cloud Engineer — Platform.sh

January 2021 – October 2021 • Remote

Contributed to core platform infrastructure for 10K+ applications. Improved observability and performance. Led the Python 3 core components migration.

Key Achievements:

- Designed observability features with Prometheus and custom metrics exporters, improving platform visibility and cutting debug time by 50%
- Led the Python 2 to Python 3 migration for 15+ critical services and libraries, preserving compatibility and improving performance by 25%
- Standardized coding conventions and established linting and testing frameworks across 20+ repositories, cutting code review time by 40%
- Partnered with Operations and Support to cut P1 incident resolution time by 35% through better monitoring and runbooks

Technologies: Python, Go, SQL, LXC

Lead Software Engineer — Clevernet

January 2015 – January 2021 • Barcelona, Spain

First senior engineering hire. Established engineering practices and platform architecture for an early-stage networking startup.

Key Achievements:

- Built an SD-WAN platform from scratch that supported 500+ enterprise deployments across 20+ countries
- Designed and implemented an IPsec/MQTT distributed control plane. Reached 99.9% uptime and sub-50ms failover between WAN connections
- Set up DevOps practices with CI/CD, Ansible IaC, and monitoring, cutting deployment time from days to hours
- Grew and mentored the engineering team from 2 to 8 engineers. Set code review processes, testing standards, and architecture documentation
- Evaluated and chose core platform technologies for scale, security, and operations

Technologies: Python, Ansible, MQTT, IPsec, PostgreSQL, Linux networking, SD-WAN, AWS

Software Engineer — Careesma

December 2012 – January 2015 • Barcelona, Spain

Contributed to a job board platform with 6+ million unique visitors and 1+ million emails sent per month across India, Austria, and Poland.

Key Achievements:

- Ported a legacy PHP/C++ application to Python and simplified jobseeker onboarding

Technologies: Python, Solr, MySQL

Software Engineer — Qustodio (Evolium spin-off)

January 2011 – November 2012 • Barcelona, Spain

Joined the initial engineering team for parental control and internet monitoring software.

Key Responsibilities:

- Built data gathering and processing (ETL) systems
- Built a REST API for the parent dashboard and desktop application
- Implemented alerts and notifications service

Technologies: Python, PHP, Hadoop, MySQL, AWS

Software Engineer — Evolium

November 2008 – November 2012 • Barcelona, Spain

Built security solutions for a well-known antivirus software company.

Key Projects:

- Built a client library that communicated with cloud protection and security services
- Built a phishing and malware protection cloud service with heuristic threat detection
- Built a web app to administer antivirus installations remotely

Technologies: C++, C#, MySQL

Research Assistant — Universitat Politècnica de Catalunya

January 2007 – November 2008 • Barcelona, Spain

Conducted research in the Broadband Communications Research Group (CBA) on broadband networks and services.

Key Projects:

- Built network QoS measurement tools: NetMeter and OreNETa
- Contributed to the EU EuQoS project

Technologies: C++, Qt, XML/XSLT, QoS

Education

MSc in Computer Science
Universitat Politècnica de Catalunya • 2003 – 2008
Telematic Networks and Operating Systems

Languages

English	Professional	**Catalan**	Native
Spanish	Native	**Italian**	Basic